

Linking Traps to Dielectric Breakdown through Charge Dynamics for Polymer Nanocomposites

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ABSTRACT

Polymer nanocomposites can change the density and/or energy of traps, suppress the accumulation of space charges, and enhance dielectric breakdown strength. It is of interest to reveal the influencing mechanism of trap properties on the dielectric breakdown of polymer nanocomposites. Results of thermally stimulated depolarization current and surface potential decay were reviewed, showing that incorporating a small amount of nanoparticles into a polymer can increase the density and/or energy of deep traps. Then, the relation between traps and dc breakdown field of several polymer nanocomposites were analyzed. It was found that the increase in the density and energy of deep traps contributes to the improved dielectric breakdown performance. The modifications of traps by nanoparticles and surface treatments affect the charge dynamics in the bulk of polymer nanocomposites. Then, the accumulation of space charges, the distortion of electric field, and the energy gain of free carriers are regulated to improve the performance of dielectric breakdown.

Index Terms - Charge dynamics, dielectric breakdown, polymer nanocomposite, trap.

1 INTRODUCTION

CHARGE dynamics includes charge transport, energy gain and dissipation, impact ionization and charge multiplication, etc., and determines insulation failure of dielectric materials. Charge transport properties of polymeric insulating materials determine their space charge accumulation behavior, which can distort electric field [1-3]. When the distorted electric field exceeds a threshold value, impact ionization, current multiplication, electromechanical forces, and energy transfer processes associated with charge detrapping can initiate aging, deterioration or dielectric breakdown in the insulating materials [3-7], reducing the lifespan of power equipment and spacecraft. However, the relation between charge transport and dielectric breakdown are not fully clear until now.

Charge transport is influenced by traps in insulating materials. Traps can be generated by physical and chemical defects [8, 9]. It is usually assumed that charges in shallow traps produced mainly by physical defects [8] are in thermodynamic equilibrium in the time window of experiments such as space charge or transient space charge limited current measurements. In other words, the maximum time constant of charge trapping and detrapping dynamics in shallow traps is much smaller than the response time of experimental

apparatuses. We can just consider the charge migration, but neglect the charge trapping and detrapping dynamics in shallow traps. In addition, chemical impurities can form deep traps [9]. It will take very long time for the charge trapping and detrapping dynamics of deep traps to reach an equilibrium state, which is confirmed by space charge measurements [10-13]. Therefore, the charge trapping and detrapping dynamics of deep traps should be considered during the analysis of transient electrical properties, and dielectric breakdown of polymeric insulating materials.

In recent years, polymer nanocomposites prepared by incorporating nanofillers into polymers have attracted great attention in the fields of dielectrics and electrical insulation [14-19]. Polymer nanocomposites exhibit many superior electrical properties such as higher breakdown strength, less space charge accumulation, higher threshold voltage for partial discharge, and less electrical tree formation [7, 14-19].

The relation between breakdown strength and nanoparticle loading were obtained by analyzing experimental results of several kinds of polymer nanocomposites [7]. A ratio k defined as the breakdown strength of polymer nanocomposites divided by that of the polymer matrix is used to evaluate their breakdown performance. Various kinds of polymer nanocomposites such as alumina-based polyimide (PI/Al₂O₃), barium titanate-based polyvinyl acetate (PVA/BaTiO₃), silver-based epoxy resin (EP/Ag), and titania-based polyimide

(PI/TiO₂) nanocomposites show enhanced breakdown strength at relatively low loadings of nanoparticles. It indicates that the breakdown strength increases at low loadings, even in the case of metal nanoparticles [7].

Nanofillers change traps in the bulk leading to the change in charge transport characteristics such as carrier mobility and conductivity. These modifications regulate the distributions of space charges and electric field in a polymeric material, and reduce the energy gain of free carriers, contributing to the improvement of dielectric breakdown [7, 14-19].

It seems that deep traps play an important role in the processes of charge transport and dielectric breakdown. In the present work, we investigate the relation between deep traps and dielectric breakdown of polymeric insulating materials by considering charge dynamics influenced by trap parameters.

2 TRAPS IN POLYMER NANOCOMPOSITES

2.1 TRAPS IN A POLYMERIC MATERIAL

Based on energy band theory, localized states or traps in a polymeric material are illustrated in Figure 1 [3, 20, 21]. Polymeric insulating materials have a large band gap (about 7-9 eV). Physical and chemical defects can create localized states or traps. Physical defects may produce localized states near the edges of both conduction and valence bands (extended states for electrons and holes). These localized states correspond to shallow traps [8]. Chemical defects may introduce deep traps, which are located away from the edges [9]. Generally, the density of shallow traps is larger than that of deep traps, and the density may decrease gradually with an increase in trap energy. Accordingly, the inter-trap distance increases with increasing trap energy. In addition, the retention time of trapped charges is smaller in shallow traps than in deep traps. The increases in the inter-trap distance and the retention time of trapped charges result in the mobility of carriers through hopping in traps decreasing dramatically with an increase in trap energy. Therefore, we can obtain a mobility edge for electrons or for holes. We also can use the mobility edges to separate shallow and deep traps in the forbidden gap [3, 20, 21].

2.2 INFLUENCE OF INTERFACIAL REGIONS ON TRAPS IN A POLYMER NANOCOMPOSITE

The addition of nanoparticles can affect the structure of a polymeric material. The Small Scale Effect of nanoparticles

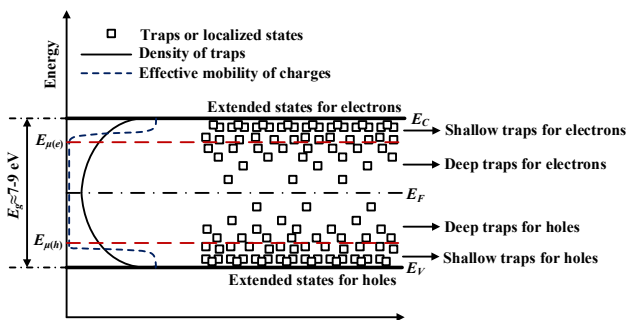


Figure 1. Schematic of trap distribution in energy band of a polymeric material; E_C : extended states for electrons, E_V : extended states for holes, E_F : Fermi level, E_g : forbidden band gap, $E_{\mu(e)}$: mobility edge of electrons, $E_{\mu(h)}$: mobility edge of holes.

and their large specific surface area may contribute to the variations of molecular chain movements (or conformation), crystallization behavior (or aggregation structure), and morphology of polymer nanocomposites. The change of conformation depends largely on interfacial regions between nanoparticles and polymer matrix, leading to the modification of traps, charge dynamics, and breakdown characteristics [7, 14-17, 19, 22, 23].

Several models such as Lewis' model [14, 15], Tanaka's model [16, 17], and the model proposed by us [7] have been used to interpret the changes in dielectric properties of polymer nanocomposites. For example, in the last model [7], an interfacial region is divided into three subregions, namely, bonded, transitional, and normal region. They contribute to the changes in different aspects of material properties. A bonded region is connected to a nanoparticle by bonds such as covalent, ionic, and hydrogen bonds. It can introduce deep traps in polymer nanocomposites. A transitional region is considered as a crystalline region and can influence the cohesive energy density, crystallinity, glass transition temperature, and chain mobility, shallow traps, and so on. In a normal region, molecular chains arrange randomly as those in the polymer matrix [7]. Accordingly, the superior space charge suppression and breakdown properties may be derived from the role of interfacial regions [7, 14-17, 19].

3 TRAP PROPERTIES OF POLYMER NANOCOMPOSITES

Figure 2 shows experimental results of thermally stimulated depolarization current (TSDC) of alumina-based low-density polyethylene (LDPE/Al₂O₃) nanocomposites performed in a TSDC system manufactured by novocontrol technologies GmbH & Co. KG [24]. Their thicknesses d are about 200 μm . The loadings of nano-Al₂O₃ are 0.1 wt%, 0.5 wt%, 1 wt%, 2 wt%, 5 wt%, and 10 wt%. The samples were polarized at a voltage of 250 V at 353 K ($\approx 80^\circ\text{C}$) for 15 minutes. Then, the samples were cooled down to a low temperature T_0 by liquid nitrogen, which was 173 K ($\approx -100^\circ\text{C}$). The short-circuit currents were measured by a picoammeter (6517B, Keithley) during heating processes with a rising rate of $3.3 \times 10^{-2} \text{ K s}^{-1}$.

The energy and density of traps can be obtained by fitting experimental results of TSDC to the following equation [25]:

$$j(T) = B \exp \left[-\frac{E_T}{k_B T} - \frac{1}{\xi \tau_0} \int_{T_0}^T \exp \left(-\frac{E_T}{k_B T'} \right) dT' \right] \quad (1)$$

where $j(T)$ is the TSDC current density in A m^{-2} , B is a constant in A m^{-2} , E_T is the activation energy of the relaxation process in eV, ξ is the heating rate in K s^{-1} , τ_0 is the relaxation time constant in s, k_B is the Boltzmann constant, and T is the temperature in K.

The energy and density of traps in LDPE/Al₂O₃ nanocomposites were obtained by fitting the experimental results to equation (1). It was found that both the energy and density of deep traps as shown by peak α in Figure 2 increase firstly and then decrease with nanoparticle loading [24]. Compared with the results reported previously [26, 27], it is concluded that

peak α is contributable to trapped charges. In the present work, we discuss the relation between the density and energy of deep traps and breakdown strength, so we focus on the trap parameters of peak α .

The densities of trapped charges in LDPE/ Al_2O_3 nanocomposites were in the order of 10^{16} m^{-3} , which is much smaller than the density of traps [24]. This may be caused by the low voltage applied to the samples during polarization process. The density of trapped charges n_0 increases with an increase in the applied electric field F_{appl} before most of the traps are filled [27]. According to the space charge limited current theory [28–30], the maximum space charges $Q_{\text{sc(max)}}$ accumulated in a dielectric material is smaller than the production of permittivity $\epsilon_0\epsilon_r$ and applied electric field F_{appl} , namely $\epsilon_0\epsilon_r F_{\text{appl}}$. Here, ϵ_0 is the vacuum permittivity in Fm^{-1} , while ϵ_r is the dielectric constant of a dielectric material. The dielectric constants of LDPE nanocomposites measured by dielectric relaxation spectrometer (DRS, novocontrol concept 80) were around 2.3–2.6 at 50 Hz [31]. Since an electric field of $1.25 \times 10^6 \text{ Vm}^{-1}$ was applied on LDPE nanocomposites in the TSDC experiments, we can estimate that the maximum space charges $Q_{\text{sc(max)}}$ accumulated inside is smaller than $2.88 \times 10^{-5} \text{ Cm}^{-2}$ with ϵ_r of 2.6 (for LDPE/ Al_2O_3 nanocomposite with the loading of 10wt%) [31]. The density of trapped charges n_0 at temperature T_0 after being cooled by liquid nitrogen can be evaluated by the equation $n_0 = Q_{\text{sc(max)}}/e$ [27]. Here, e is the elementary charge in C. Then, we can obtain that n_0 is smaller than $8.98 \times 10^{17} \text{ m}^{-3}$. In addition, some of the injected charges might be captured by traps with lower energy such as relaxation peak β . Therefore, it may be reasonable that the densities of deep traps evaluated from the relaxation peak α are in the order of 10^{16} m^{-3} .

We also used surface potential decay method (SPD) to characterize the trap properties of polymer nanocomposites. Neat EP and alumina-based epoxy resin (EP/ Al_2O_3) nanocomposite was charged by a corona discharging with a needle-grid electrode system at room temperature for 2 min. The radius of the needle tip was about 10 μm . The distance between the needle and grid was about 10 mm, and the distance between the grid and sample was also about 10 mm. The grid and needle voltages were -4 and -9 kV, respectively. After charging, the surface potential ϕ_s was measured as a

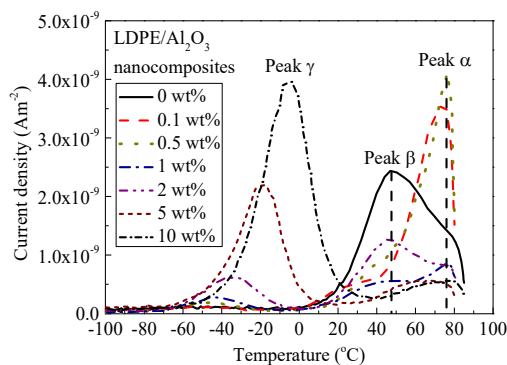


Figure 2. Experimental results of thermally stimulated current of 200- μm thick LDPE/ Al_2O_3 nanocomposites. The loadings of nano- Al_2O_3 are listed in the figure. The samples were polarized at a voltage of 250 V at 353 K for 15 min. The currents were measured by a picoammeter (6517B, Keithley) during heating processes with a rising rate of $3.3 \times 10^{-2} \text{ Ks}^{-1}$ [22].

function of time t at a constant temperature by a non-contacting voltmeter (P0865, Trek). The experimental results were analyzed by Simmon's theory [32–34]. According to Simmon's theory [32], the energy of traps can be calculated by the following equation:

$$E_T = k_B T \ln(\nu_{\text{ATE}} t). \quad (2)$$

Here, ν_{ATE} is the attempt-to-escape frequency in s^{-1} . The density of traps is proportional to the products of time t and decay rates of surface potential, namely, $t d\phi_s/dt$. Figure 3 shows the energy and density distributions of neat EP and alumina-based epoxy resin (EP/ Al_2O_3) nanocomposite calculated by Simmon's theory. It is shown that the density of deep traps is larger in EP/ Al_2O_3 nanocomposite than in neat EP, while the energy of deep traps is nearly the same.

4 LINKING TRAPS TO DC BREAKDOWN

4.1 RELATION BETWEEN TRAP PROPERTIES AND DC BREAKDOWN

It has been found that incorporating a small amount of nanoparticles (e.g., < about 2wt%) into polymers can increase the energy or density of deep traps [35–37]. TSDC results of polymer nanocomposites indicated that the energy and density of traps were changed by nanofillers. The deep trap energy and density initially increased and then decreased with increasing nanofiller loading in LDPE/ Al_2O_3 nanocomposites [35]. The temperature of TSDC peak shifted to high temperature in PI/silica nanocomposites with 2 wt% and 5 wt% nanosilica [38], indicating deeper traps are introduced by incorporating nanofillers. Some other results also verified that the deep traps were introduced in nanocomposites such as silica-based cross-linked polyethylene (XLPE/ SiO_2) [36], zinc oxide-based low-density polyethylene (LDPE/ ZnO) [39], and PI/ TiO_2 [40] nanocomposites. It was also found that polymer nanocomposites present better suppression for space charge accumulation and higher breakdown strength, compared to neat polymers and microcomposites [11, 19, 36, 41]. It was

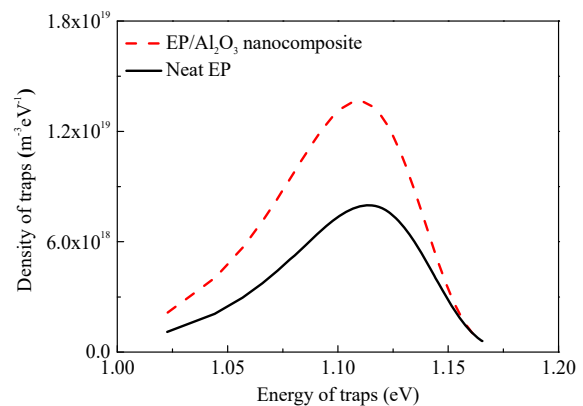


Figure 3. Energy and density distributions of neat EP and EP/ Al_2O_3 nanocomposite. The loading of nano- Al_2O_3 was 2wt%. The thicknesses of the samples are about 1 mm. The samples were charged by corona discharging with a needle-grid electrode system at room temperature for 2 min. Both the distance between the needle and the grid and that between the grid and the sample were about 10 mm. The radius of the needle tip was about 10 μm . The grid and needle voltages were -4 and -9 kV, respectively. The surface potential was measured by a non-contacting voltmeter (P0865, Trek).

suggested that the bonded and transitional regions determine the short-term breakdown and long-term breakdown properties, respectively [7]. Since the bonded region can increase the energy or density of deep traps, the property of deep traps may have a strong relation with breakdown.

In order to find the relation between traps and breakdown performance of polymer nanocomposites, the typical data about the energy and density of traps and breakdown strength from many papers are analyzed. A breakdown ratio k_1 defined as the breakdown field of a polymer nanocomposite divided by that of its neat polymer is used to evaluate the breakdown performance similarly to our previous paper [7].

Figure 4 shows the relation between the energy of deep traps and breakdown field F_b of polymer nanocomposites. The trap energies of LDPE/ Al_2O_3 and XLPE/ SiO_2 nanocomposites were characterized by TSDC method [35, 36]. The thicknesses d of LDPE/ Al_2O_3 nanocomposites used in dc breakdown tests are about 200 μm . Two spherical electrodes with a diameter of 25 mm were used in dc breakdown tests. All the breakdown tests were carried out in transformer oil at room temperature. A ramp dc voltage with a rising rate V_{ramp} of 500 Vs^{-1} was applied on the samples. The breakdown field of neat LDPE samples $F_{b(\text{neat})}$ is about $2.6 \times 10^8 \text{ Vm}^{-1}$ [35]. For XLPE/ SiO_2 nanocomposites, d , V_{ramp} , and $F_{b(\text{neat})}$ are equal to 15–150 μm , 500 Vs^{-1} , and $2.7 \times 10^8 \text{ Vm}^{-1}$ (at 298 K) [36]. Recessed specimens with gold electrodes were used in the dc breakdown experiments of XLPE/ SiO_2 nanocomposites [36]. For both LDPE/ Al_2O_3 and XLPE/ SiO_2 nanocomposites, the breakdown field increases with an increase in the energy of deep traps.

Figures 5 shows the relation between the energy of deep traps and breakdown ratio k_1 . Here, we assume that the energy of deep traps is independent of electric field and temperature, namely, it is not changed during the breakdown experiment. The breakdown ratio k_1 increases with an increase in the energy of deep traps. In LDPE/ Al_2O_3 nanocomposites, the energy of deep traps increases obviously, resulting in higher breakdown strength [35]. Additionally, through the modification of nanoparticle surface, the trap parameters are also changed. For example, the trap energy increases in XLPE hexamethyldisilazane (HMDS) and n-(2-aminoethyl) 3-aminopropyltrimethoxysilane (AEAPS) treated silica nanocomposites. The polar surface groups introduced by these surface modifiers would create surface states, forming deep traps [36]. Charge carriers injected from electrodes can be captured by deep traps near the interfaces between the dielectric material and its electrodes, reducing the interfacial electric field. The subsequent injection currents will be suppressed, and the distortion of electric field will be mitigated. In addition, an increase in trap energy provide a higher potential barrier for mobile charge carriers. They may need gain more energy from the applied electric field to overcome the potential barrier. Consequently, the breakdown strength increases as shown in Figures 4 and 5 [35, 36].

As mentioned above, we can obtain trapped charges in deep traps by TSDC. Since trapped charges reflects the density of traps, we used the trapped charges in deep traps to analyze the relation between the density of deep traps and breakdown performance. Figures 6 and 7 show the influence of trapped

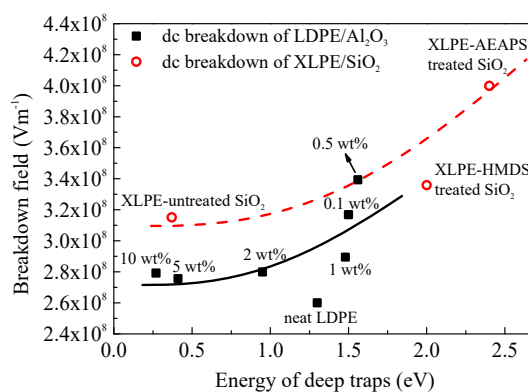


Figure 4. Relation between the energy of deep traps and breakdown field of polymer nanocomposites. ■: data from [32] and ○: data from [33].

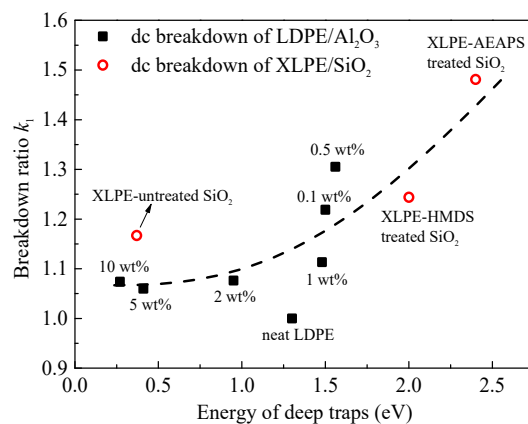


Figure 5. Relation between the energy of deep traps and breakdown ratio k_1 of polymer nanocomposites. ■: data from [32] and ○: data from [33]. For LDPE/ Al_2O_3 , the trap energy and density were obtained by TSDC measurement (polarized at 2 kV/mm at 343 K); for XLPE/silica the trap level was obtained by TSC measurement (polarized at 5 kV/mm). For LDPE/ Al_2O_3 nanocomposites, $d=200 \mu\text{m}$, $V_{\text{ramp}}=500 \text{ Vs}^{-1}$, and $F_{b(\text{neat})}=2.6 \times 10^8 \text{ Vm}^{-1}$. For XLPE/ SiO_2 nanocomposites, $d=15\text{--}150 \mu\text{m}$, $V_{\text{ramp}}=500 \text{ Vs}^{-1}$, and $F_{b(\text{neat})}=2.7 \times 10^8 \text{ Vm}^{-1}$.

charges in deep traps on the breakdown field and breakdown ratio k_1 of LDPE/ Al_2O_3 nanocomposites, respectively [35]. The breakdown strength increases with an increase in trap density of LDPE/ Al_2O_3 nanocomposites [35]. It indicates that an increase in the density of deep traps corresponds to higher breakdown strength of nanocomposites at lower nanoparticle loading.

Figures 4 to 7 show that both increasing the energy and density of deep traps may be beneficial for the breakdown performance. It is suggested that the increase in deep traps depends on the role of interfacial regions [7, 18, 19]. The variation of local morphology such as free volume and chain movements caused by interfaces may play an important role in the increase in deep traps. Deep traps can capture mobile charge carriers. It is hard for charges to detrapp from deep traps. This can reduce the mobility and density of charge carriers. According to the breakdown mechanisms, the energy obtained by charge carriers from applied field would be reduced. The polymer chain fracture may become difficult, causing higher breakdown field.

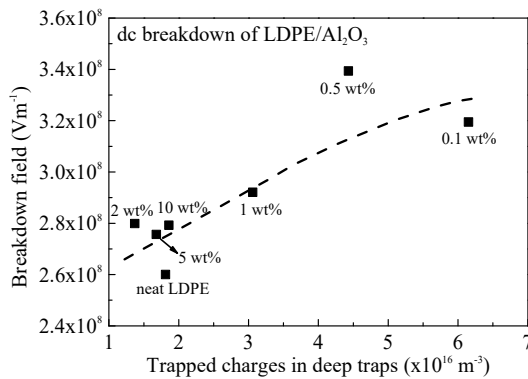


Figure 6. Relation between the density of deep traps and breakdown ratio k_1 in nanocomposites. ■: data from [32].

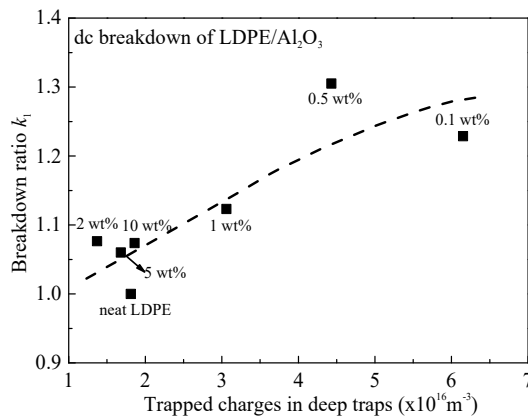


Figure 7. Relation between the density of deep traps and breakdown ratio k_1 in nanocomposites. ■: data from [32]. Here, $d=200\ \mu\text{m}$, $V_{\text{ramp}}=500\ \text{Vs}^{-1}$, and $F_{b(\text{neat})}=2.6\times 10^8\ \text{Vm}^{-1}$.

4.2 RELATION BETWEEN CARRIER MOBILITY AND DC BREAKDOWN

Carrier mobility is a micro parameter of charge transport that is closely related to traps. Shallow traps have little effects on carrier mobility, while deep traps can decrease it significantly. Usually, an effective carrier mobility is used to evaluate the transport of charges. The modification of traps (particularly deep traps) by nanofillers changes carrier mobility, leading to the change in breakdown strength. For instance, the carrier mobility of XLPE/SiO₂ nanocomposites was estimated using a method based on absorption current measurements [36]. The carrier mobility was calculated by $\mu=0.786d^2/t_pV$. Here, t_p is the time at which the absorption current is its maximum [36].

The result indicated that the carrier mobility increases with an increase in electric field in neat XLPE and its microcomposites. However, the carrier mobility exhibits a decreasing trend with an increase in electric field in nanocomposites, and is lower than that of neat XLPE and its microcomposites [36]. The carrier mobility in LDPE/MgO nanocomposites was estimated by dividing the velocities of charge carriers calculated from space charge profiles by the applied voltage of $20\ \text{kVmm}^{-1}$. The results demonstrate that the carrier mobility decreases with increasing the loading of MgO nanofillers. The reductions in carrier mobility are caused

by the trapping effect of interfaces between polymer and nanofillers. The trapping effect suppresses charge injections from electrodes and hinders the motion of charge carriers in the bulk of nanocomposites [41].

Figures 8 and 9 show the dependences of breakdown field and breakdown ratio k_1 on carrier mobility of nanocomposites [36, 41]. For LDPE/MgO nanocomposites, d , V_{ramp} , and $F_{b(\text{neat})}$ are equal to $100\ \mu\text{m}$, $500\ \text{Vs}^{-1}$, and $6.2\times 10^8\ \text{Vm}^{-1}$ [41]. The dc breakdown experiments of LDPE/MgO nanocomposites were carried out with McKeown-type electrodes in silicone oil at $303\ \text{K}$ [41]. It is obvious that lower carrier mobilities lead to higher breakdown strength. In LDPE/MgO nanocomposites, the carrier mobility can be reduced to $1.2\times 10^{-14}\ \text{m}^2\text{V}^{-1}\text{s}^{-1}$ in nanocomposites with 1 phr nano-MgO from $4.1\times 10^{-14}\ \text{m}^2\text{V}^{-1}\text{s}^{-1}$ in neat LDPE [41]. For XLPE/SiO₂ nanocomposites, surface modifications of nano-SiO₂ significantly decrease the charge mobility. The carrier mobility of triethoxyvinylsilane (TES) surface treatment nanocomposites is $8.0\times 10^{-15}\ \text{m}^2\text{V}^{-1}\text{s}^{-1}$, which nearly decreases by two orders of magnitude compared with that of neat XLPE ($3.7\times 10^{-13}\ \text{m}^2\text{V}^{-1}\text{s}^{-1}$), and it exhibits the highest breakdown strength [36] as shown in Figures 8 and 9.

4.3 RELATION BETWEEN CONDUCTIVITY AND DC BREAKDOWN

Electrical conductivity depends on the mobility and density of charge carriers. These two parameters can be influenced by traps, causing different conductivities. It should be noted that dc conductivity is a macro parameter related to traps. In general, an insulating material with a low conductivity leads to a high breakdown strength. DC conductivities of nanocomposites were studied, and various results show that dc conductivity decreases after an addition of nanofillers. For example, the dc conductivity of LDPE/Al₂O₃ nanocomposites is lower than that of neat LDPE, and it initially decreases and then increases with an increase in nanofiller loadings [24]. Incorporation of nano-MgO can obviously decrease the dc conductivity of LDPE [41]. The same reduction in dc conductivity is also observed in silica-based polyimide (PI/SiO₂) nanocomposites [38]. These reductions in dc conductivity can be ascribed to deep traps introduced by nanofillers. The results are consistent with the carrier mobility as shown in Figure 9. The dc conductivities of LDPE/Al₂O₃, LDPE/MgO, and PI/SiO₂ were measured under electric fields of 40, 40, and 5 kVmm^{-1} , respectively.

Figures 10 and 11 show the relations between high field dc conductivity and breakdown field and between dc conductivity and breakdown ratio k_1 of nanocomposites, respectively. For PI/SiO₂ nanocomposites, d , V_{ramp} , and $F_{b(\text{neat})}$ are equal to $25\ \mu\text{m}$, $500\ \text{Vs}^{-1}$, and $1.5\times 10^8\ \text{Vm}^{-1}$, respectively [38]. It can be seen that the reduction in dc conductivity gives rise to the increase in breakdown strength of nanocomposites such as the 2 wt% LDPE/MgO and 10 wt% PI/SiO₂ nanocomposites. The decrease in dc conductivity indicates that space charge behavior is improved, resulting in the enhancement of breakdown performance.

4.4 DISCUSSION

Polymers have different states in different ranges of temperatures according to the motion of molecular chains [42]. They are in glass-like state below glass transition temperature,

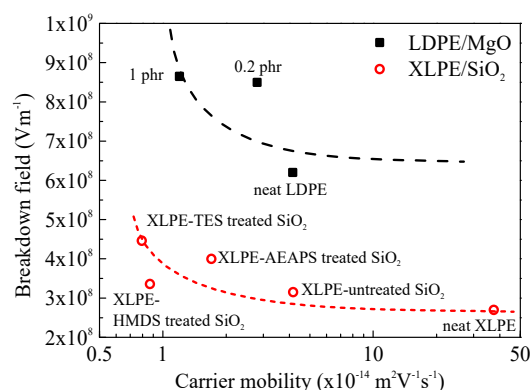


Figure 8. Breakdown field as a function of charge mobility in nanocomposites. ■: data from [38] and ○: data from [33].

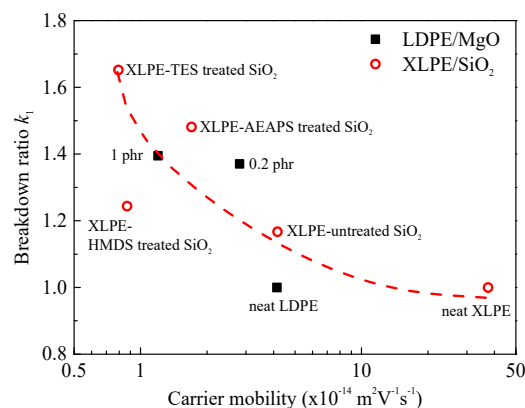


Figure 9. Breakdown ratio k_1 as a function of charge mobility in nanocomposites. ■: data from [38] and ○: data from [33]. For LDPE/MgO nanocomposites, $d=100\ \mu\text{m}$, $V_{\text{ramp}}=500\ \text{Vs}^{-1}$, and $F_{b(\text{neat})}=6.2\times10^8\ \text{Vm}^{-1}$. For XLPE/SiO₂ nanocomposites, $d=15\text{--}150\ \mu\text{m}$, $V_{\text{ramp}}=500\ \text{Vs}^{-1}$, and $F_{b(\text{neat})}=2.7\times10^8\ \text{Vm}^{-1}$.

in rubber-like state between glass transition and melting temperatures, and in plastic-flow state above melting temperature [42]. The breakdown mechanisms are different in these three states as shown in Figure 12 [3-5]. In glass-like state, the breakdown field of polymers is very high and independent of temperature or increase slightly with a decrease in temperature. The breakdown process can be explained by models of electron avalanche breakdown, intrinsic breakdown, or field emission breakdown [3-5]. In rubber-like state, the breakdown field decreases with an increase in temperature. The breakdown behavior is influenced by many effects such as multiplication of electrons, current, or thermal power dissipation [2, 3, 43], energy gain in free volume [44], enhancement of mechanical forces [6, 45]. The breakdown process can be interpreted by models of collective breakdown [4, 5], thermal breakdown [4, 5, 46], free volume breakdown [44], electromechanical breakdown [6], and filamentary electromechanical breakdown [45]. It is also found that secondary effects such as space charges and molecular displacement affect the breakdown process [3-5, 7, 10, 47-49]. In plastic-flow state, the breakdown is mainly caused by thermal or electromechanical process.

Space charge and electric field distributions in polyethylene at high electric fields demonstrate that the distortion of electric

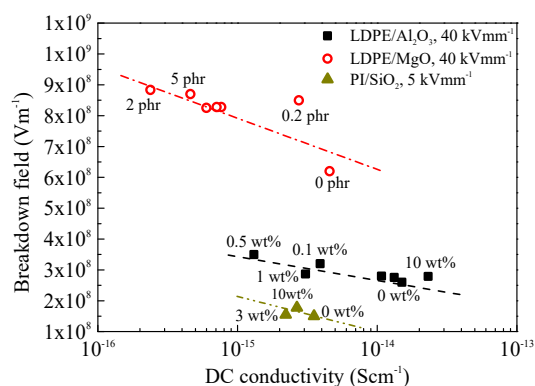


Figure 10. Relation between high field dc conductivity and breakdown ratio k_1 in nanocomposites. ■: data and from [32], ○: data from [38], and ▲: data from [35].

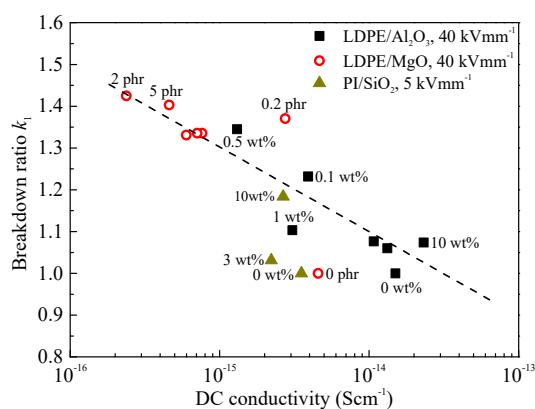


Figure 11. Relation between high field dc conductivity and breakdown ratio k_1 in nanocomposites. ■: data and from [32], ○: data from [38], and ▲: data from [35]. For LDPE/Al₂O₃ nanocomposites, $d=200\ \mu\text{m}$, $V_{\text{ramp}}=500\ \text{Vs}^{-1}$, and $F_{b(\text{neat})}=2.6\times10^8\ \text{Vm}^{-1}$. For LDPE/MgO nanocomposites, $d=100\ \mu\text{m}$, $V_{\text{ramp}}=500\ \text{Vs}^{-1}$, and $F_{b(\text{neat})}=6.2\times10^8\ \text{Vm}^{-1}$. For PI/SiO₂ nanocomposites, $d=25\ \mu\text{m}$, $V_{\text{ramp}}=500\ \text{Vs}^{-1}$, and $F_{b(\text{neat})}=1.5\times10^8\ \text{Vm}^{-1}$.

field caused by the accumulation of space charges may determine the breakdown field. It was found that the maximum electric field before breakdown is twice as high as the applied electric field on LDPE sample with a thickness of $70\ \mu\text{m}$ by pulsed electroacoustic apparatus [11], and is 2.5 times for the sample with a thickness of $130\ \mu\text{m}$ [10]. In addition, it was found that the maximum electric field before breakdown is 8 times higher than the applied electric field on $13.5\ \text{mm}$ -thick XLPE sample measured by pressure wave propagation method [49]. Therefore, it seems that the breakdown process of polyethylene may be controlled by secondary effect of space charges.

Interfacial regions between nanoparticles and polymer matrix can change the local structures of polymer nanocomposites, causing the change in the density and energy of traps [7, 16, 17, 35, 50]. It may reduce carrier mobility of polymer nanocomposites [18, 36] and suppress charge injection into and space charge accumulation in the nanocomposites [11, 19, 41, 51, 52]. Since deep traps can capture charge carriers and reduce their mobility, they affect charge transport in insulating materials [53]. Space charges are

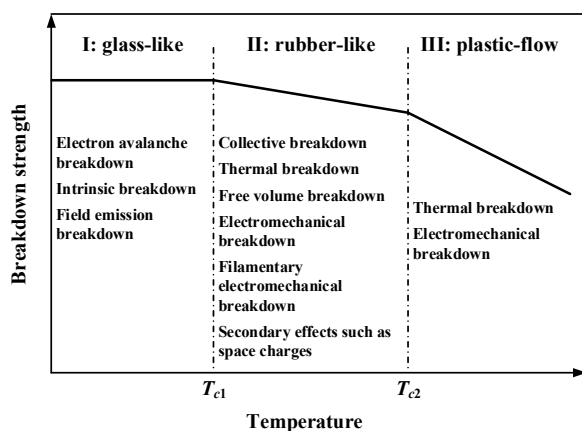


Figure 12. Schematic of breakdown strength of polymeric insulating materials as a function of temperature [3-5].

accumulated in polymers under the application of voltages. According to Poisson's equation, the more space charges are accumulated, the severer electric field will be distorted. The increases in the energy and/or density of deep traps can reduce the effective carrier mobility and the mean-free-path of free carriers. The charge injection can be suppressed because of the reduction in the interfacial electric field, leading to less accumulation of space charges and weaker distortion of electric field. In addition, free carriers can gain less energy from the electric field as the mean-free-path and the local electric field decrease. Then, the breakdown field will be increased. Spatial distributions of space charges and electric field before breakdown show that the density of space charges are much lower and the distortion of electric field is much smaller in LDPE/MgO nanocomposites than in neat LDPE [10, 11]. Accordingly, the breakdown field is higher for LDPE/MgO nanocomposites than for neat LDPE. It was also found that breakdown is triggered at the position where the electric field is the highest [10, 11].

Furthermore, the interfacial regions can also be considered as scattering centers for carriers and can reduce the energy of mobile charges [16, 17]. The diffuse layers of mobile charges create local conductivity, reducing the charge accumulation [7, 54]. All the changes in the energy and density of traps and carrier mobility, the formation of scattering centers, and the creation of local conductivity may affect the breakdown behavior of polymer nanocomposites.

5 CONCLUSION

The conclusions drawn from the present research can be summarized as follows:

(1) Trap properties of polymer nanocomposites was measured by TSDC and SPD. It was found that the energy or density of traps is changed with an increase in loading of nanoparticles.

(2) The density and energy of deep traps, carrier mobility, dc conductivity, and dc breakdown field of several polymer nanocomposites were analyzed to reveal the relation between traps and dc breakdown performance. It was found that the dc breakdown field increases with an increase in the density and energy of deep traps and with a decrease in carrier mobility and dc conductivity.

(3) The breakdown process of polyethylene nanocomposites may be controlled by secondary effect of space charges. The change in energy or density can influence charge transport and electric field distortion in nanocomposites, affecting the breakdown behavior.

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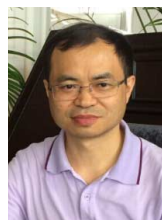
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